

④

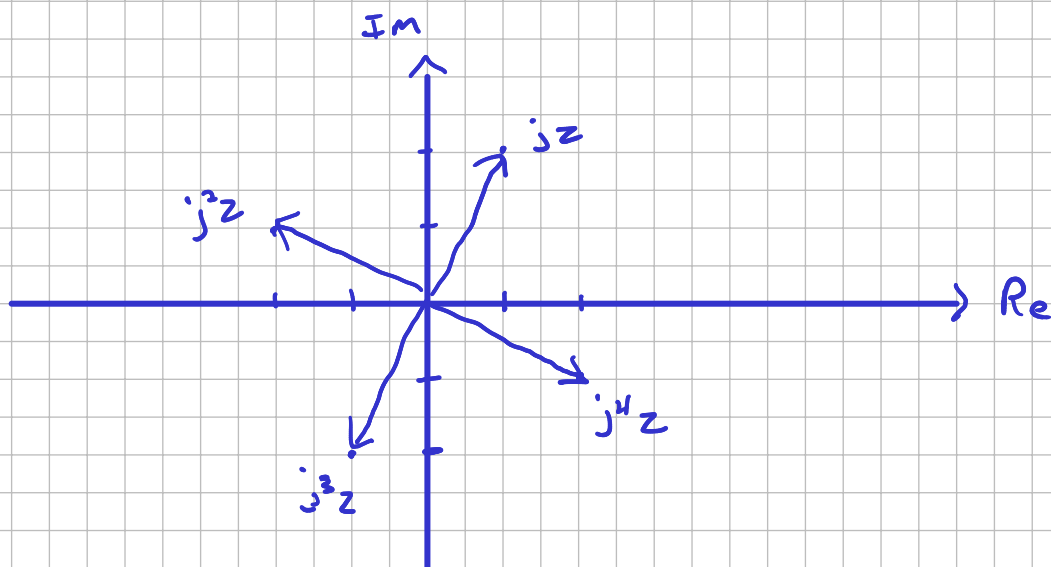
Let $z = 2 - j$.compute jz, j^2z, j^3z, j^4z .

$$\begin{aligned}
 jz &= j(2 - j) = 2j - j^2 \\
 &= 2j - (-1) \\
 &= \underline{1 + 2j} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 j^2z &= (-1)z = (-1)(2 - j) \\
 &= \underline{-2 + j} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 j^3z &= (-j)(2 - j) = -2j + j^2 \\
 &= \underline{-1 - 2j} \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 j^4z &= (j^2)(j^2)(2 - j) = (-1)(-1)(2 - j) \\
 &= 1z = \underline{2 - j} \quad \checkmark
 \end{aligned}$$

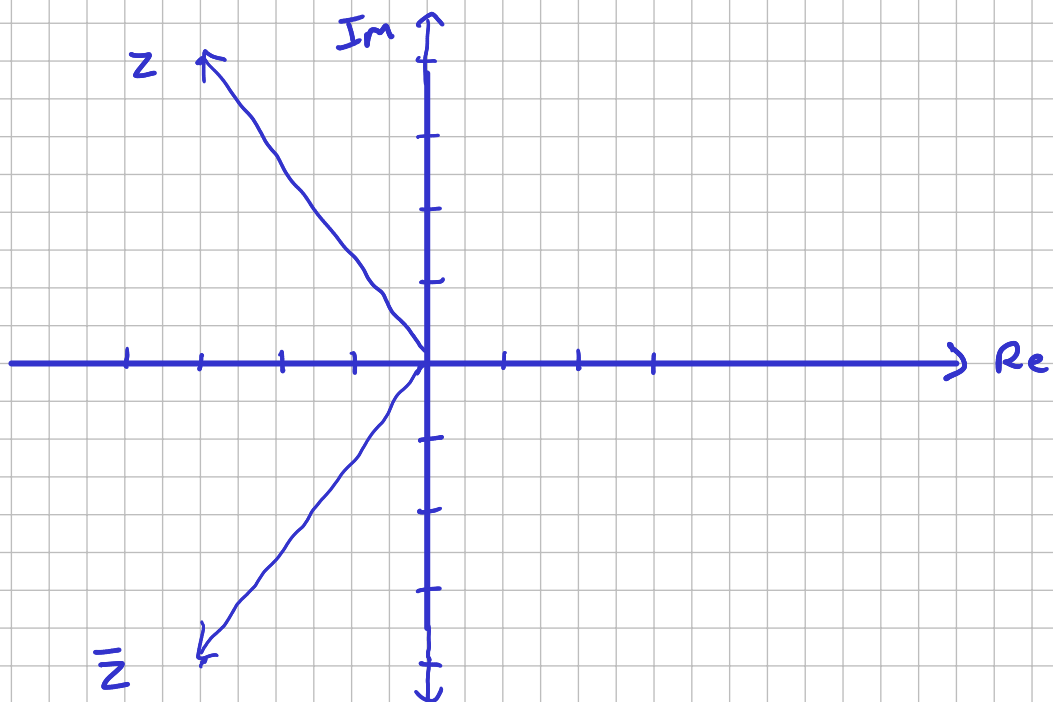


⑤ For $z = -3 + 4j$
determine $\bar{z}$, $|z|$, $z\bar{z}$.

- $\bar{z} = -3 - 4j$

- $|z| = |-3 + 4j|$
 $= \sqrt{(-3)^2 + (4)^2}$
 $= \sqrt{9 + 16} = 5$

- $z\bar{z} = |z|^2 = 5^2 = 25$



⑥ Write $\frac{3-2j}{1+3j}$ in rectangular form: $a + bj$

$$\Rightarrow \frac{3-2j}{1+3j} \cdot \frac{1-3j}{1-3j}$$

$$\begin{aligned}\text{Numerator: } (3-2j)(1-3j) &= 3 - 9j - 2j + 6j^2 \\ &= 3 - 11j - 6 \\ &= \underline{-3 - 11j}\end{aligned}$$

$$\begin{aligned}\text{Denominator: } (1+3j)(1-3j) &= 1 - 3j + 3j - 9j^2 \\ &= 1 - 9j^2 \\ &= 1 + 9 = \underline{10} \quad \checkmark\end{aligned}$$

$$\text{Rectangular } \Rightarrow \frac{-3 - 11j}{10} = \underline{\underline{\frac{-3}{10} - \frac{11j}{10}}} \quad \checkmark$$

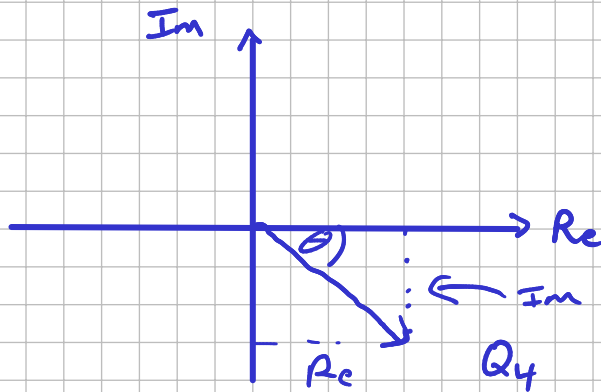
⑧ Convert the following to rectangular form:

a) $z = 4e^{-j\pi/3}$

b) $w = 5[\cos(150^\circ) + j\sin(150^\circ)]$

Given: moduli (4 and 5) & arguments ($-\pi/3$ and 150°)

a) $r = 4, \theta = -\frac{\pi}{3}$



$$\sin \theta = \frac{\text{Im}}{r}$$

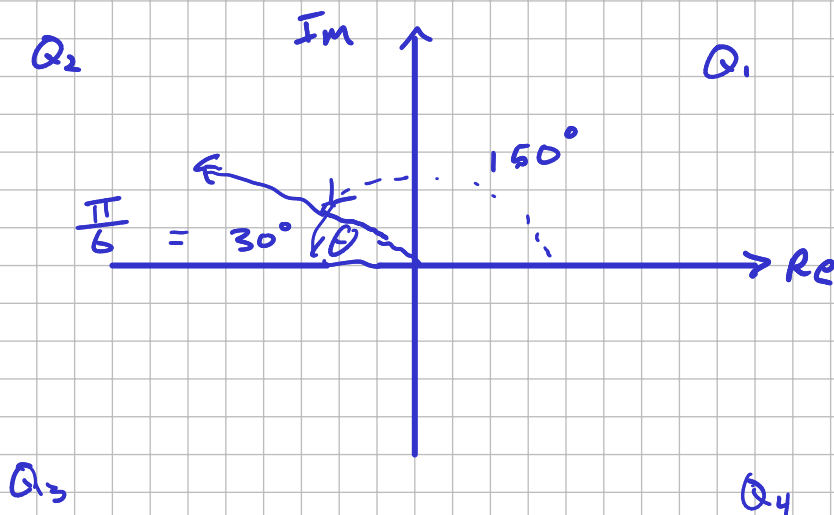
$$\cos \theta = \frac{\text{Re}}{r}$$

$$\begin{aligned}\text{Im}(z) &= r \sin \theta = 4 \sin(-\pi/3) \\ &= -4(\sqrt{3}/2) = \underline{-2\sqrt{3}}\end{aligned}$$

$$\text{Re}(z) = r \cos \theta = 4(1/2) = \underline{2} \quad \checkmark$$

Rectangular form: $z = \underline{2 - 2\sqrt{3}j} \quad \checkmark$

b) $w = 5[\cos(150^\circ) + j\sin(150^\circ)]$



$$|w| = 5$$

$$\theta = \frac{\pi}{6} \quad |_{Q_2}$$

$$|w| = 5$$

$$\theta = \frac{\pi}{6} \Big|_{Q_2}$$

$$\sin \theta = \frac{Im}{|w|} \Rightarrow |w| \sin(\pi/6)$$

$$Im(w) = 5 \left(\frac{1}{2} \right) = \underline{\underline{\frac{5}{2}}} \quad \checkmark$$

$$\cos \theta = \frac{Re}{|w|} \Rightarrow Re(w) = |w| \cos(\pi/6)$$

$$= 5 \cos(\pi/6)$$

Rectangular Form:

$$-5 \frac{\sqrt{3}}{2} + \frac{5}{2} j$$

minus
b/c Q_2 !

$$= -5 \left(\frac{\sqrt{3}}{2} \right)$$

$$(II) \quad x(t) = 5 \cos(\omega t - 30^\circ)$$

→ A phasor is a "snapshot" of a sinusoidal signal in time

→ Its purpose is to "freeze" a signal to move it from the time domain t into the freq. domain!

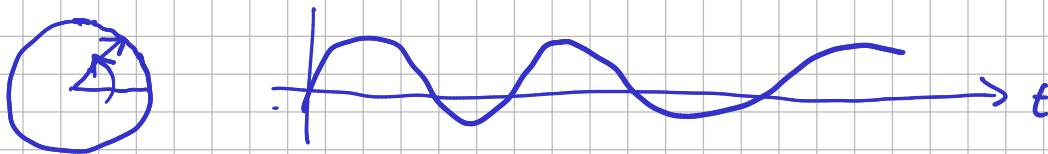
$$\boxed{X e^{j\omega t} \Rightarrow X \angle \theta} \Rightarrow j\omega t = j\theta$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

$$\Rightarrow \omega t = \frac{2\pi}{T} t = \theta$$

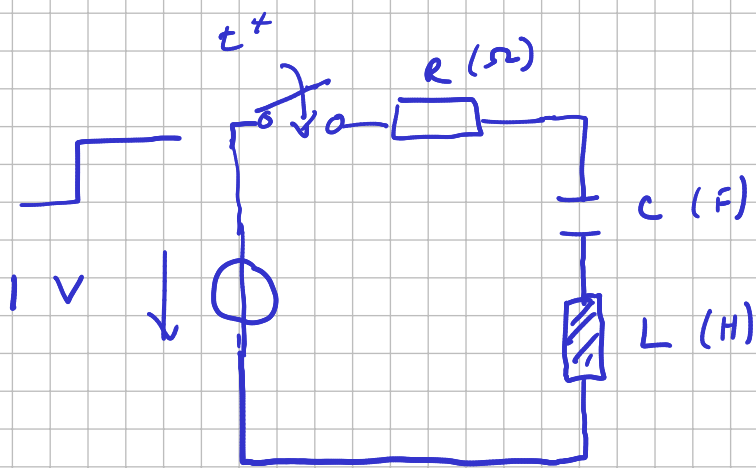
$$\Rightarrow \text{Phasor: } 5 \angle -30^\circ$$

→ This is for steady-state oscillatory signals/systems

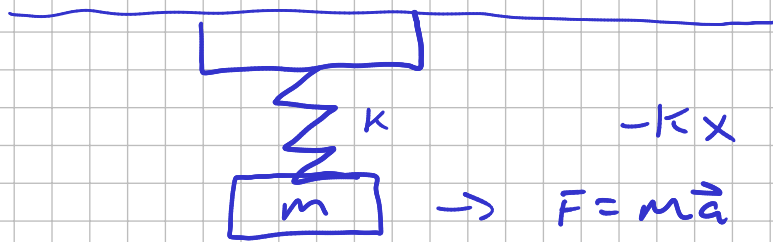


Steady rotation

Transient: Damped oscillatory system $e^{j(\omega - \omega_0)t}$



Mass - Spring



$$-kx$$

$$\rightarrow F = m\vec{a}$$

2nd order